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Feather corticosterone is lower in translocated and historical populations of the endangered Laysan duck (*Anas laysanensis*)

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Identifying reliable bioindicators of population status is a central goal of conservation physiology. Physiological stress measures are often used as metrics of individual health and can assist in managing endangered species if linked to fitness traits. We analysed feather corticosterone, a cumulative physiological stress metric, of individuals from historical, translocated, and source populations of an endangered endemic Hawaiian bird, the Laysan duck (*Anas laysanensis*). We hypothesized that feather corticosterone would reflect the improved reproduction and survival rates observed in populations translocated to Midway and Kure Atolls from Laysan Island. We also predicted less physiological stress in historical Laysan birds collected before ecological conditions deteriorated and the population bottleneck. All hypotheses were supported: we found lower feather corticosterone in the translocated populations and historical samples than in those from recent Laysan samples. This suggests that current Laysan birds are experiencing greater physiological stress than historical Laysan and recently translocated birds. Our initial analysis suggests that feather corticosterone may be an indicator of population status and could be used as a non-invasive physiological monitoring tool for this species with further validation. Furthermore, these preliminary results, combined with published demographic data, suggest that current Laysan conditions may not be optimal for this species.

1. Introduction

The use of physiological measures to assess and monitor the health of a population in the face of conservation challenges has become a valuable technique [1–3]. It has increased in popularity as the field of conservation physiology has grown. This approach can be particularly beneficial for monitoring and protecting endangered species if the biomarkers selected can be linked reliably to fitness outcomes [4–7]. Several types of physiological indicators, including immunological, bioenergetic and endocrine tools, have been employed to understand how changes in habitat quality affect population status and dynamics [8–10]. Glucocorticoids (GCs), a component of the physiological stress response, are often used due to their role in responding to and recovering from environmental stressors and their links to individual fitness [11,12]. Brief elevations in GCs following a perturbation are typically beneficial and help to restore homeostasis; however, repeated or prolonged elevations can lead to wear and tear on the organism and pathological effects [13]. GCs have been used to investigate the impacts of anthropogenic disturbance and changes in habitat quality in mammals [14,15], birds [16–19], reptiles [20–22], amphibians [23–26] and fish [27–29], and they have been linked to changes in demography (reviewed in [6]).

Blood samples are the most commonly used method for measuring GC concentrations, but several other methods that span various time scales have been developed over the past few decades [30]. Keratinized tissues, such as feathers, offer a cumulative measure of circulating GC concentrations over the weeks during their growth [30–35]. Feather corticosterone (cort) is sensitive to stressors lasting a week to a month, making it suitable for tracking ongoing, chronically stressful conditions during the period of feather growth [30]. It has been used to investigate the effects of habitat fragmentation [19], pollution [36,37], prey quality [38], differences in body condition [36,37,39,40], survival in captivity [39,41] and other metrics of fitness [42]. Collecting feathers to measure cort is a minimally to non-invasive technique that can be accomplished without capture. Moulded feathers can be collected passively, or feathers can be taken from deceased individuals [33]. Due to its non-invasive and integrative nature, feather cort is an attractive biomarker for monitoring endangered species (e.g. [40]).

We used feather cort as a physiological biomarker to assess the status of an endemic, endangered Hawaiian bird, the Laysan duck (*Anas laysanensis*). This species has a highly restricted range and is listed as endangered by the U.S. Fish and Wildlife Service (USFWS) and critically endangered by the International Union for Conservation of Nature (IUCN). Once widespread throughout the Hawaiian archipelago, human colonization and its associated impacts restricted their range to Lisianski and Laysan Islands [43–45]. They were extirpated on Lisianski around 1840 [46,47] and thereafter persisted only on Laysan. Their population continued to decrease due to hunting by guano miners from 1891 to 1903, the introduction of rabbits around 1903 and hunting by illegal feather harvesters from 1908 to 1910 [48]. The rabbits removed almost all of the island's vegetation, eliminating nesting habitat and cover and allowing drifting sand to partially fill the lagoon and several freshwater seeps [48,49]. In 1902, there were fewer than 100 individuals [50], and by 1912, the population had decreased to just seven adults and five juveniles [51] (figure 1). Three of the five endemic bird species found on Laysan went extinct during this time, including the Laysan rail (*Porzana palmeri*), Laysan millerbird (*Acrocephalus familiaris*) and Laysan honeycreeper (*Himatione sanguinea freethii*) [48]. Rabbits were eradicated from Laysan in 1923, and the population began to recover [48]. Over the past century, the population size has undergone several significant fluctuations. After recovering to an estimate of 688 individuals in 1961 [52,53], it dipped to fewer than 100 in 1993 [47], only to rebound to more than 500 by 2004 [54] (figure 1).

As part of the proposed recovery for this species [53], in 2004–2005, 42 Laysan ducks were translocated to Midway Atoll to establish a second population [54,55]. Following release, the Midway population demonstrated high survival [55] and novel reproductive traits that increased fertility and, in turn, fitness. Females on Midway bred at an earlier age—less than 1 year of age in some instances—while those from Laysan did not breed until 2 years of age. On Midway, females were more likely to renest following clutch failure, raise two successful broods and even make three or more nesting attempts in a season. With these broods, Midway females laid larger clutches—almost double the size of those on Laysan—and could raise up to 11 ducklings in a single brood [59,60]. While adult and post-fledging juvenile survival rates were similar to those on Laysan [55,56,61], the chick survival rate was 63% on Midway [55] compared with 28% on Laysan [62]. Laysan's most recent population size estimate is 265–413 individuals [57,58], while on Midway, it is 920–1054 individuals [58].

In 2014, 28 individuals were translocated from Midway to Kure Atoll, where they demonstrated similar survival rates to those seen on Midway [56] (figure 1). The population is currently 87 individuals and persisting there and on Midway despite chronic avian botulism outbreaks [56,63]. A second translocation is being developed, and additional monitoring of reproductive behaviour on Kure is planned [58]. The higher chick survival rates, increased fertility and novel life history traits demonstrated by the translocated populations at Midway, and the similar adult survival rates on Kure and Midway [56] provide an opportunity to link physiological variation to observed changes in fitness. We investigated an integrated, retrospective measure of stress, feather cort, to determine whether it covaried with the observed differences in demography between Laysan, Kure and Midway. We hypothesized that the translocated populations' improved environmental conditions would be associated with reduced physiological stress.

Finally, because feather cort is stable and can be measured in museum specimens [36,38,64–66], we also investigated feather cort from Laysan duck specimens collected prior to the ecological devastation and population bottleneck caused by the rabbit introduction and duck harvest to determine whether it differs from feather cort concentrations in more recent samples. Our prediction was that with the altered vegetation, erosion, continued drifting of sand and the resulting reduction in access to freshwater through the filling of seeps [53,58], Laysan Island has become a harsher, less suitable environment for this species, increasing their physiological stress.

2. Methods

(a) Sample collection

Feather samples were collected from naturally deceased individuals found on Midway during regular monitoring. Samples from Kure and Laysan were taken from specimens at the Museum of Wildlife and Fish Biology at the University of California, Davis, and the Bishop Museum in Honolulu, HI. All samples were shipped to Tufts University, Medford, MA, for analysis. Samples from Laysan were grouped according to the date of specimen collection for the historical (1903) versus contemporary (1978–2007) analysis (table 1). Additional details about the samples can be found in the electronic supplementary material, table S1. In Laysan ducks, moult occurs between June and August and takes 3–4 weeks to complete [47]; thus, the cort contained in the feathers analysed should represent the conditions experienced and related physiological stress over about a month at a similar time of year across samples.

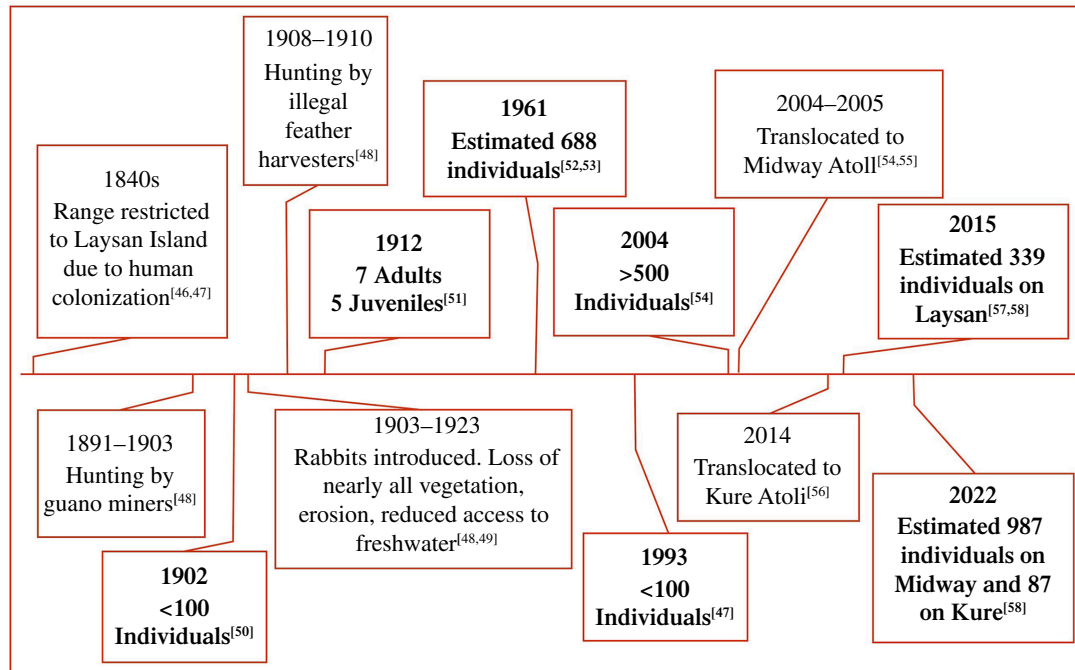


Figure 1. Timeline of the exploitation, near extinction, recovery and translocation of the Laysan duck. Bold text indicates population estimates. Superscript numbers indicate reference numbers [46–58].

(b) Extraction and assay

Feather samples were prepared and assayed following the methods of Bortolotti *et al.* [31] and Lattin *et al.* [32], with some modifications. Samples collected were a mixture of flight and body feathers. For each feather, the calamus was removed, then the feather was measured to the nearest millimetre to calculate a total length, cut into sections of 5 mm² or less and placed into a 15 ml Falcon tube. Sample masses were standardized to ≥25 mg to avoid the nonlinear relationship between sample mass and the concentration of corticosterone detected [32]. The distal portion was used first if only a partial feather section was needed.

To begin the extraction, 7 ml of methanol were added to the Falcon tubes containing the minced feathers. Samples were placed into a sonicating water bath at room temperature for 30 minutes, then incubated overnight in a shaking water bath at 50°C. Feathers were separated from methanol using vacuum filtration through #4 Whatman filters. Feather remnants, filter and sample tube were washed twice with 2.5 ml of methanol. Filtered methanol was then evaporated under nitrogen gas in a 50°C water bath. Dried steroid extracts were then reconstituted in 500 µl of assay buffer (X065 Buffer, Arbor Assays, Ann Arbor, MI, USA). Samples were covered and stored at 4°C until analysis.

Feather cort concentrations were analysed using a commercially available enzyme immunoassay kit (cat. #K014, Arbor Assays, Ann Arbor, MI, USA) according to the manufacturer's instructions. This kit has been used previously to analyse feather corticosterone successfully [37,67,68]. All samples and standards were run in duplicate, and feather pools were included on both plates. Duplicates were averaged to obtain reported concentrations. The inter-assay coefficient of variation was 6.97%, and the intra-assay coefficient of variation was 8.87%.

(c) Statistical analysis

All statistical analyses were completed using R v.4.3.1 [69] in RStudio v.20203.6.1.438 [70]. Feather cort was log-transformed to normalize the data for analysis (Shapiro–Wilk test; $W = 0.98$, $p = 0.76$). We compared mean feather cort among Laysan-Historical, Laysan-Contemporary, Kure and Midway samples. We used the 'leveneTest' function (car package [71]) to test for homogeneity of variance ($F_{3,35} = 1.05$, $p = 0.38$), then used a one-way analysis of variance using the 'aov' function. We applied *post hoc* Fisher's least significant difference tests with the Benjamini–Hochberg p -adjustment using the 'LSD.test' function (agricolae package [72]) to check for pairwise differences in mean feather cort.

3. Results

Mean feather cort differed significantly among the samples from Laysan-Historical, Laysan-Contemporary, Midway and Kure (figure 2; $F_{3,35} = 7.285$, $p < 0.001$). Mean feather cort concentrations were statistically significantly lower in Laysan-Historical ($\bar{X} \pm \text{s.d.}$: 5.62 ± 2.14 pg mg⁻¹; $p < 0.05$), Midway ($\bar{X} \pm \text{s.d.}$: 4.70 ± 2.38 pg mg⁻¹; $p < 0.01$) and Kure samples ($\bar{X} \pm \text{s.d.}$: 3.60 ± 2.24 pg mg⁻¹; $p < 0.001$) compared with Laysan-Contemporary samples ($\bar{X} \pm \text{s.d.}$: 9.61 ± 4.31 pg mg⁻¹). Midway samples were not statistically significantly different from Kure samples ($p = 0.18$) or Laysan-Historical samples ($p = 0.28$). Mean feather cort was significantly lower in Kure samples than in historical Laysan samples ($p < 0.05$).

Table 1. Summary of the locations, sample sizes and dates of collection of the feather samples analysed. For detailed information, see the electronic supplementary material, table S1.

location	sample size	years collected
Kure	$n = 8$	2016–2021
Midway	$n = 12$	2021–2022
Laysan-Contemporary	$n = 9$	1978–2007
Laysan-Historical	$n = 10$	1903

4. Discussion

As we predicted, feather cort was lower in the samples from the translocated populations compared with the source population on Laysan Island, indicating that the individuals in the translocated populations were experiencing less physiological stress. Taken with the improvements in fertility and survival, this result suggests that the translocation improved conditions for this species. Additionally, feather cort from samples taken before the habitat degradation of Laysan Island and near extinction of the Laysan duck over a century ago were lower than those taken more recently from Laysan Island birds. This suggests that some of the damage done to the island has effects on Laysan ducks that persist to the present day.

The lower feather cort concentrations in the samples from Midway and Kure imply that the individuals in the translocated populations are experiencing less stress than those from the source population. This could be explained by greater habitat quality in the new locations. While Laysan is approximately 415 ha in size [73], over half of that area is occupied by unvegetated blowout, beach and dunes, and by a 74 ha hypersaline lagoon [74]. Midway Atoll is larger, at approximately 600 ha, and provides denser and more diverse vegetation and approximately 2.2 ha of freshwater wetlands [75]. Midway also offers a greater diversity of invertebrates for prey [76]. Restoration efforts on Midway prior to the translocation included planting native sedges and bunchgrasses, which provide nesting habitat [55] and excavation of nine freshwater seeps [54]. We propose that the increased access to freshwater and more abundant prey could have driven these results. Increases in circulating cort concentration can occur as a result of dehydration [77,78] and poor food availability [79–82]. Additionally, hypersaline water, as found on Laysan, can be toxic to ducklings [83] and fatal for waterfowl of all age classes [84,85]. Together, these characteristics support the proposition that Midway is a better site for Laysan ducks.

It is possible that the concentrations of cort measured in the feathers from populations on Midway and Kure could have an interpretation that is the reverse of what we suggest: that low values were due to greater chronic stress. Dickens and Romero [86] reviewed the literature on chronic stress effects on circulating (blood) cort levels in terrestrial vertebrates and found no consistent patterns; chronic stress in some species results in elevated baseline cort values, while in other species it results in low baseline values. These circulating levels would drive levels in feather samples. The definitive test of whether low cort values indicate good health or chronic stress requires an animal in-hand, to determine if it can mount a physiological stress response [87,88]. In the absence of this in-hand physiological ‘challenge’, one needs to rely on additional information to interpret the comparatively low cort values we documented. From studies of other bird species, we know that translocation—capture, captivity and transportation—is stressful, and cort concentrations are typically elevated upon release due to novelty and uncertainty in the new environment [89–94]. If the stressors are persistent or prolonged, symptoms of chronic stress may arise. For example, Dickens *et al.* [95] found that the process of translocation can have additive effects that lead to lower baseline cort, reduced ability to mount an acute stress response and impaired negative feedback (see also [96]). It is important to recall that the stress response is part of a healthy individual’s physiology, but chronic stress is not [97]. All the specimens were collected 2 or more years after translocation occurred, and some birds were likely the descendants of translocated individuals. So, to distinguish low cort values as indicators of no versus chronic stress, we would look for additional indicators, such as lower survival and reproduction in the translocated population [89,96,98,99]. Instead, similar levels of adult survival [56] and significant improvements in fertility, juvenile survival and earlier breeding were reported [60]. The evidence, therefore, suggests that the introduced populations recovered well and flourished in their new environment, which suggests that the observed low levels of feather cort cannot be attributed to the effects of chronic stress.

When comparing feather cort concentration over time on Laysan, we find support for the hypothesis that habitat quality has decreased on Laysan over time. The concentrations in historical samples, taken before the rabbit introduction and the associated habitat degradation, are significantly lower than those from samples taken more recently. Historically, the ducks experienced less physiological stress than they currently do on Laysan. Additionally, the historical feather cort concentrations are not significantly different from those measured in the samples from Midway, where they appear to be flourishing. Although the population size on Laysan has recovered, the physiological markers of the population suggest that Laysan is no longer optimal habitat for this species. The continued filling of freshwater seeps by drifting sand is likely a contributing factor [58]. Thus, despite ongoing restoration efforts, Laysan may no longer be the best home for its namesake duck, and additional translocations are likely to benefit the species. However, we note that the sample size for this comparison is necessarily limited due to the origins of the samples. Additional samples collected throughout the past 120 years would provide more context on how physiological stress has changed over time as habitat quality on Laysan has changed.

The concentrations of feather cort measured in the source and translocated Midway populations aligned well with the observed demographic differences and improved fertility. This may indicate that feather cort could be a good bioindicator for this species. However, we must note that this is just an observational association and not a demonstrated causal relationship.

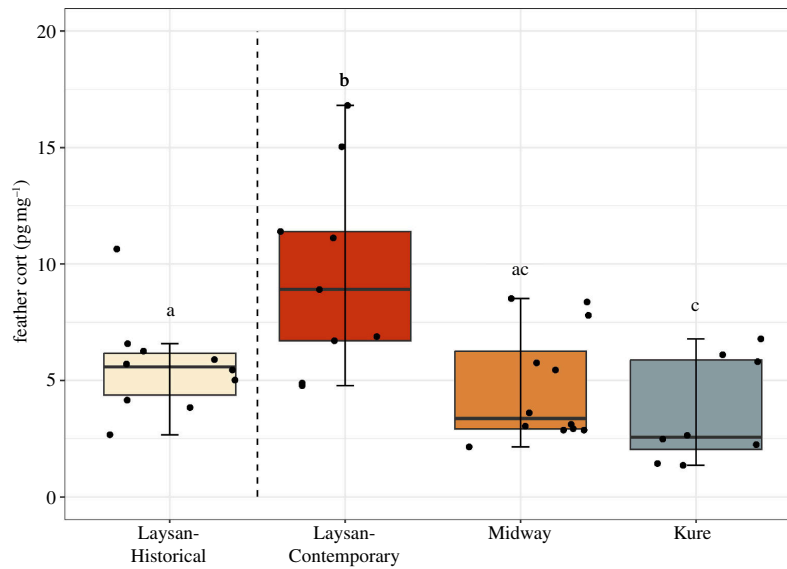


Figure 2. Feather cort concentrations are different among the Laysan-Historical, Laysan-Contemporary, Midway and Kure samples. Letters indicate statistically significant differences from pairwise comparisons ($p < 0.05$). Bold midline indicates median, boxes represent 1st and 3rd quartiles and whiskers indicate maximum and minimum. Points outside of whiskers are outliers. Untransformed data are shown here but log-transformed data were used for the analysis and pairwise comparisons.

The increased fertility observed in the Midway ducks may be a result of increased reproductive effort in response to the improved habitat quality. Baseline cort is typically elevated in the breeding season [100] and increased reproductive effort can lead to increased circulating cort levels—that would be detected in feather cort [30]—to help meet elevated metabolic demands [101,102]. Yet even with these potential elevations, the feather cort levels in the Midway samples are still significantly lower than the levels in the current Laysan samples. Therefore, it is likely that even if breeding effort has increased, the overall physiological stress that they are experiencing has decreased. Prior work has found mixed support for the relationships between feather cort and reproduction, environmental conditions or body condition. Harris *et al.* [42] saw no relation between feather cort and an experimental increase in past reproductive investment or reproduction under natural conditions that followed. Conversely, Crossin *et al.* [103] found that feather cort was related to both breeding success and subsequent breeding attempts, but found higher feather cort in successful breeders suggesting an energetic trade-off. Alternatively, feather cort may reflect environmental conditions rather than reproductive investment [104]. Some studies reported no relationship between alterations to habitat and feather cort [40], while other studies report associations between past disturbances and elevated feather cort [105]. Finally, feather cort does reflect body condition and survival in some contexts [39], but does not correlate with body condition in others [40]. Consequently, there is high variation across studies, so the potential usefulness and accuracy of feather cort as a biomarker must be evaluated for each species it is used with. For the Laysan duck, the concordance between the changes in life history and feather cort between the Laysan and Midway populations support the idea of a relationship, but manipulative experiments would be needed to support this conclusion. If feather cort is a reliable biomarker, we can use it as an indicator of how the translocated population on Kure is performing. As the concentrations found in the Kure samples are not significantly different from those in the Midway samples, this suggests that they are experiencing similarly lower physiological stress levels. Additionally, the concentrations in the Kure samples are slightly lower than those of the historical Laysan samples. It is plausible that their performance is comparable to that of the ducks in the Midway population and perhaps even slightly improved compared with the Laysan population of 100 years ago. However, it is important to bear in mind that there are several covariates and confounding factors in each of these comparisons that cannot be disentangled within this analysis due to its unique structure and sample sources. Additional tests of this biomarker in future studies of this species will provide insight into its potential usefulness.

5. Conclusions

In a recent paper, Cooke *et al.* [7] highlight several critical questions in conservation physiology for threatened species, including: (1) What physiological information is most helpful to policy makers in determining the status of populations, species or individuals? and (2) Which physiological tools can we use to improve the health, condition and fitness of endangered species? In this work, we have begun to tackle these questions for the endangered Laysan duck. Feather cort appears to be a valuable non-invasive physiological monitoring tool for this species due to its apparent association with fecundity. However, additional demographic data and experimental evidence, such as an ACTH challenge to assess individual capacity to mount a stress response [87,88] are needed to determine the degree to which feather cort is a reliable bioindicator of population health for this species [86,106]. Nevertheless, our initial analysis suggests that this tool has the potential to provide information on population status and could be used to assess outcomes from conservation efforts such as habitat restoration or translocations in the future. Our preliminary physiological results and published demographic data suggest that current conditions on Laysan

Island are no longer optimal for its namesake duck. However, the USFWS' efforts to conserve this species using translocations have been effective.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. Data and code are available through Dryad [107].

Electronic supplementary material is available online [108].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. D.A.V.K.: conceptualization, data curation, formal analysis, investigation, methodology, visualization, writing—original draft, writing—review and editing; L.M.R.: conceptualization, funding acquisition, project administration, writing—original draft, writing—review and editing; J.M.R.: conceptualization, project administration, resources, writing—original draft, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

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References

- Wikelski M, Cooke SJ. 2006 Conservation physiology. *Trends Ecol. Evol.* **21**, 38–46. (doi:10.1016/j.tree.2005.10.018)
- Madliger CL *et al.* 2016 Success stories and emerging themes in conservation physiology. *Conserv. Physiol.* **4**, cov057. (doi:10.1093/conphys/cov057)
- Madliger CL, Franklin CE, Love OP, Cooke SJ. 2021 *Conservation physiology: applications for wildlife conservation and management*. Oxford, UK: Oxford University Press. (doi:10.1093/oso/9780198843610.001.0001)
- Birnie-Gauvin K, Walton S, Palme CAD, Manouchehri BA, Venne S, Lennox RJ, Chapman JM, Bennett JR, Cooke SJ. 2017 Conservation physiology can inform threat assessment and recovery planning processes for threatened species. *Endang. Species Res.* **32**, 507–513. (doi:10.3354/esr00831)
- Mahoney JL, Klug PE, Reed WL. 2018 An assessment of the US endangered species act recovery plans: using physiology to support conservation. *Conserv. Physiol.* **6**, coy036. (doi:10.1093/conphys/coy036)
- Bergman JN, Bennett JR, Binley AD, Cooke SJ, Fyson V, Hlina BL, Reid CH, Vala MA, Madliger CL. 2019 Scaling from individual physiological measures to population-level demographic change: case studies and future directions for conservation management. *Biol. Conserv.* **238**, 108242. (doi:10.1016/j.biocon.2019.108242)
- Cooke SJ *et al.* 2021 One hundred research questions in conservation physiology for generating actionable evidence to inform conservation policy and practice. *Conserv. Physiol.* **9**, coab009. (doi:10.1093/conphys/coab009)
- Homyack JA. 2010 Evaluating habitat quality of vertebrates using conservation physiology tools. *Wildlife Res.* **37**, 332. (doi:10.1071/WR08093)
- Albano N. 2012 Conservation physiology tools: their role in assessing habitat quality in birds. *Ardeola* **59**, 197–216. (doi:10.13157/arla.59.2.2012.197)
- Madliger CL, Love OP, Hultine KR, Cooke SJ. 2018 The conservation physiology toolbox: status and opportunities. *Conserv. Physiol.* **6**, coy029. (doi:10.1093/conphys/coy029)
- Dantzer B, Fletcher QE, Boonstra R, Sheriff MJ. 2014 Measures of physiological stress: a transparent or opaque window into the status, management and conservation of species? *Conserv. Physiol.* **2**, cou023. (doi:10.1093/conphys/cou023)
- McCormick SD, Romero LM. 2017 Conservation endocrinology. *BioScience* **67**, 429–442. (doi:10.1093/biosci/bix026)
- Romero LM, Wingfield JC. 2016 *Tempests, poxes, predators, and people: stress in wild animals and how they cope*. New York, NY: Oxford University Press. (doi:10.1093/acprof:oso/9780195366693.001.0001)
- Balestri M, Barresi M, Campera M, Serra V, Ramanamanjato JB, Heistermann M, Donati G. 2014 Habitat degradation and seasonality affect physiological stress levels of *Eulemur collaris* in littoral forest fragments. *PLoS One* **9**, e107698. (doi:10.1371/journal.pone.0107698)
- Kaisin O, Fuzessy L, Poncin P, Brotcorne F, Culot L. 2021 A meta-analysis of anthropogenic impacts on physiological stress in wild primates. *Conserv. Biol.* **35**, 101–114. (doi:10.1111/cobi.13656)
- Wasser SK, Bevis K, King G, Hanson E. 1997 Noninvasive physiological measures of disturbance in the northern spotted owl. *Conserv. Biol.* **11**, 1019–1022. (doi:10.1046/j.1523-1739.1997.96240.x)
- Thiel D, Jenni-Eiermann S, Braunisch V, Palme R, Jenni L. 2008 Ski tourism affects habitat use and evokes a physiological stress response in capercaillie *Tetrao urogallus*: a new methodological approach. *J. Appl. Ecol.* **45**, 845–853. (doi:10.1111/j.1365-2664.2008.01465.x)
- Monti F, Duriez O, Dominici J-M, Sforzi A, Robert A, Fusani L, Grémillat D. 2018 The price of success: integrative long-term study reveals ecotourism impacts on a flagship species at a UNESCO site. *Anim. Conserv.* **21**, 448–458. (doi:10.1111/acv.12407)
- Bicudo T, Anciães M, Arregui L, Gil D. 2020 Effects of forest fragmentation on feather corticosterone levels in an Amazonian Avian community. *Ardeola* **67**, 229. (doi:10.13157/arla.67.2.2020.ra1)
- Romero LM, Wikelski M. 2002 Exposure to tourism reduces stress-induced corticosterone levels in Galápagos marine iguanas. *Biol. Conserv.* **108**, 371–374. (doi:10.1016/S0006-3207(02)00128-3)
- Josserand R, Dupoué A, Agostini S, Haussy C, Le Galliard JF, Meylan S. 2017 Habitat degradation increases stress-hormone levels during the breeding season, and decreases survival and reproduction in adult common lizards. *Oecologia* **184**, 75–86. (doi:10.1007/s00442-017-3841-4)
- French SS, Webb AC, Wilcoxon TE, Iverson JB, DeNardo DF, Lewis EL, Knapp CR. 2022 Complex tourism and season interactions contribute to disparate physiologies in an endangered rock iguana. *Conserv. Physiol.* **10**, coac001. (doi:10.1093/conphys/coac001)
- Newcomb Homan R, Regosin JV, Rodrigues DM, Reed JM, Windmiller BS, Romero LM. 2003 Impacts of varying habitat quality on the physiological stress of spotted salamanders (*Ambystoma maculatum*). *Anim. Conserv.* **6**, 11–18. (doi:10.1017/S1367943003003032)

24. Hopkins WA, Durant SE. 2011 Innate immunity and stress physiology of eastern hellbenders (*Cryptobranchus alleganiensis*) from two stream reaches with differing habitat quality. *Gen. Comp. Endocrinol.* **174**, 107–115. (doi:10.1016/j.ygcen.2011.08.006)
25. Millikin AR, Woodley SK, Davis DR, Anderson JT. 2019 Habitat characteristics in created vernal pools impact spotted salamander water-borne corticosterone levels. *Wetlands* **39**, 803–814. (doi:10.1007/s13157-019-01130-5)
26. Goff CB, Walls SC, Rodriguez D, Gabor CR. 2020 Changes in physiology and microbial diversity in larval ornate chorus frogs are associated with habitat quality. *Conserv. Physiol.* **8**, coaa047. (doi:10.1093/conphys/coaa047)
27. Yin Z, Lam TJ, Sin YM. 1995 The Effects of crowding stress on the non-specific immuneresponse in fancy carp (*Cyprinus carpio* L.). *Fish Shellfish Immunol.* **5**, 519–529. (doi:10.1016/S1050-4648(95)80052-2)
28. Blevins ZW, Effert EL, Wahl DH, Suski CD. 2013 Land use drives the physiological properties of a stream fish. *Ecol. Indic.* **24**, 224–235. (doi:10.1016/j.ecolind.2012.06.016)
29. Belanger CB, Vera-Chang MN, Moon TW, Midwood JD, Suski CD, Cooke SJ. 2016 Seasonal variation in baseline and maximum whole-body glucocorticoid concentrations in a small-bodied stream fish independent of habitat quality. *Comp. Biochem. Physiol. A. Mol. Integ. Physiol.* **192**, 1–6. (doi:10.1016/j.cbpa.2015.10.027)
30. Gormally BMG, Romero LM. 2020 What are you actually measuring? A review of techniques that integrate the stress response on distinct time-scales. *Funct. Ecol.* **34**, 2030–2044. (doi:10.1111/1365-2435.13648)
31. Bortolotti GR, Marchant TA, Blas J, German T. 2008 Corticosterone in feathers is a long-term, integrated measure of avian stress physiology. *Funct. Ecol.* **22**, 494–500. (doi:10.1111/j.1365-2435.2008.01387.x)
32. Lattin CR, Reed JM, DesRochers DW, Romero LM. 2011 Elevated corticosterone in feathers correlates with corticosterone-induced decreased feather quality: a validation study. *J. Avian Biol.* **42**, 247–252. (doi:10.1111/j.1600-048X.2010.05310.x)
33. Romero LM, Fairhurst GD. 2016 Measuring corticosterone in feathers: strengths, limitations, and suggestions for the future. *Comp. Biochem. Physiol. A. Mol. Integ. Physiol.* **202**, 112–122. (doi:10.1016/j.cbpa.2016.05.002)
34. Romero LM, Beattie UK. 2022 Common myths of glucocorticoid function in ecology and conservation. *J. Exp. Zool. A. Ecol. Integr. Physiol.* **337**, 7–14. (doi:10.1002/jez.2459)
35. Jenni-Eiermann S, Helfenstein F, Vallat A, Glauser G, Jenni L. 2015 Corticosterone: effects on feather quality and deposition into feathers. *Methods Ecol. Evol.* **6**, 237–246. (doi:10.1111/2041-210X.12314)
36. Strong RJ, Pereira MG, Shore RF, Henrys PA, Pottinger TG. 2015 Feather corticosterone content in predatory birds in relation to body condition and hepatic metal concentration. *Gen. Comp. Endocrinol.* **214**, 47–55. (doi:10.1016/j.ygcen.2015.03.002)
37. Branco JM, Hingst-Zaher E, Jordan-Ward R, Dillon D, Siegrist J, Fischer JD, Schiesari L, von Hippel FA, Buck CL. 2022 Interrelationships among feather mercury content, body condition and feather corticosterone in a neotropical migratory bird, the purple martin (*Progne subis subis*). *Environ. Pollut.* **314**, 120284. (doi:10.1016/j.envpol.2022.120284)
38. Fairhurst GD, Bond AL, Hobson KA, Ronconi RA. 2015 Feather-based measures of stable isotopes and corticosterone reveal a relationship between trophic position and physiology in a pelagic seabird over a 153-year period. *Ibis* **157**, 273–283. (doi:10.1111/ibi.12232)
39. Monclús L, Carbajal A, Tallo-Parra O, Sabés-Alsina M, Darwich L, Molina-López RA, Lopez-Bejar M. 2017 Relationship between feather corticosterone and subsequent health status and survival in wild Eurasian sparrowhawk. *J. Ornithol.* **158**, 773–783. (doi:10.1007/s10336-016-1424-5)
40. Gormally BMG, van Rees CB, Bowers E, Reed JM, Romero LM. 2020 Feather corticosterone does not correlate with environmental stressors or body condition in an endangered waterbird. *Conserv. Physiol.* **8**, coaa125. (doi:10.1093/conphys/coaa125)
41. Lind MA, Hórak P, Sepp T, Meitern R. 2020 Corticosterone levels correlate in wild-grown and lab-grown feathers in greenfinches (*Carduelis chloris*) and predict behaviour and survival in captivity. *Horm. Behav.* **118**, 104642. (doi:10.1016/j.yhbeh.2019.104642)
42. Harris CM, Madliger CL, Love OP. 2017 An evaluation of feather corticosterone as a biomarker of fitness and an ecologically relevant stressor during breeding in the wild. *Oecologia* **183**, 987–996. (doi:10.1007/s00442-017-3836-1)
43. Olson SL, James HF. 1991 Descriptions of thirty-two new species of birds from the Hawaiian islands: part I. non-passeriformes. *Ornithol. Monograph.* **45**, 1–88. (doi:10.2307/40166794)
44. Cooper A, Rhymer J, James HF, Olson SL, McIntosh CE, Sorenson MD, Fleischer RC. 1996 Ancient DNA and island endemics. *Nature* **381**, 484. (doi:10.1038/381484a0)
45. Burney DA, et al. 2001 Fossil evidence for a diverse biota from kauai and its transformation since human arrival. *Ecol. Monogr.* **71**, 615–642. (doi:10.2307/3100038)
46. Olson SL, Ziegler AC. 1995 Remains of land birds from Lisianski island, with observations on the terrestrial avifauna of the northwestern Hawaiian islands. *Pacific Sci* **49**, 111–125. <http://hdl.handle.net/10125/2279>
47. Moulton DW, Marshall AP. 2020 Laysan duck (*Anas Laysanensis*). In *Birds of the World* (eds SM BillermanBK Keeney, PG Rodewald, TS Schulenberg). New York, NY: Cornell Lab of Ornithology. (doi:10.2173/bow.layduc.01)
48. Ely CA, Clapp RB. 1973 The natural history of Laysan island, northwestern Hawaiian islands. *Atoll Res. Bull.* **171**, 1–361. (doi:10.5479/si.00775630.171.1)
49. Caspers H. 1981 On the ecology of hypersaline lagoons on Laysan Atoll and Kauai Island, Hawaii, with special reference to the Laysan duck, *Anas laysanensis* Rothschild. *Hydrobiologia* **82**, 261–270. (doi:10.1007/BF00048720)
50. Fisher WK. 1903 Birds of Laysan and the Leeward islands, Hawaiian group. *Bullet. United States Fish Comm.* **23**, 767–816. <https://spo.nmfs.noaa.gov/content/birds-laysan-and-leeward-islands-hawaiian-grou>
51. Bryan WA, Dill HR. 1912 Report of an expedition to Laysan Island in 1911: under the joint 446 auspices of the United States Department of Agriculture and University of Iowa. Washington, DC: Department of Agriculture, Biological Survey. See <http://www.biodiversitylibrary.org/bibliography/62940>.
52. Warner RE. 1963 Recent history and ecology of the Laysan Duck. *Condor* **65**, 3–23. (doi:10.2307/1365134)
53. U.S. Fish and Wildlife Service. 2009 Revised recovery plan for the Laysan duck (*Anas Laysanensis*). *US Fish Wildlife Serv.* **1**, ix–114. <https://www.fws.gov/node/68174>
54. Reynolds MH, Klavitter J. 2006 Translocation of wild Laysan duck *Anas laysanensis* to establish a population at Midway Atoll national wildlife refuge, United States and US pacific possession. *Conserv. Evid.* **3**, 6–8. <https://conservationevidencejournal.com/reference/pdf/2202>
55. Reynolds MH, Seavy NE, Vekasy MS, Klavitter JL, Laniawe LP. 2008 Translocation and early post-release demography of endangered Laysan teal. *Anim. Conserv.* **11**, 160–168. (doi:10.1111/j.1469-1795.2008.00166.x)
56. Reynolds MH, Hatfield JS, Courtot KN, Vanderlip C. 2020 Survival of Laysan Teal *Anas laysanensis* differs between island populations: role of chronic avian botulism. *Wildfowl* **70**, 192–210. <https://wildfowl.wwt.org.uk/index.php/wildfowl/article/view/2727>
57. Reynolds MH, Courtot KN, Brinck KW, Rehkemper CL, Hatfield JS. 2015 Long-term monitoring of endangered Laysan Ducks: index validation and population estimates 1998–2012. *J. Fish Wildlife Manag.* **6**, 92–101. (doi:10.3996/032014-JFWM-017)
58. U.S. Fish Wildlife Service. 2022 Laysan duck (*Anas Laysanensis*) 5-year review short form summary, pp. 1–11. Honolulu, HI: U.S. Fish and Wildlife Service, Pacific Islands Fish and Wildlife Office.

59. Reynolds MH, Weiser E, Jamieson I, Hatfield JS. 2013 Demographic variation, reintroduction, and persistence of an island duck (*Anas laysanensis*): demography and reintroduction of Laysan Ducks. *J. Wildl. Manage.* **77**, 1094–1103. (doi:10.1002/jwmg.582)
60. Walters JR, Reynolds MH. 2013 Experimental reintroduction reveals novel life-history variation in Laysan ducks (*Anas laysanensis*). *Auk* **130**, 573–579. (doi:10.1525/auk.2013.13070)
61. Reynolds MH, Citta JJ. 2007 Postfledging survival of Laysan ducks. *J. Wildl. Manage.* **71**, 383–388. (doi:10.2193/2005-674)
62. Reynolds MH, Work TM. 2005 Mortality in the endangered Laysan Teal *Anas laysanensis*: conservation implications. *Wildfowl* **55**, 31–48.
63. Work TM, Klavitter JL, Reynolds MH, Blehert D. 2010 Avian botulism: a case study in translocated endangered Laysan Ducks (*Anas laysanensis*) on Midway Atoll. *J. Wildlife Dis.* **46**, 499–506. (doi:10.7589/0090-3558-46.2.499)
64. Bortolotti GR, Marchant T, Blas J, Cabezas S. 2009 Tracking stress: localisation, deposition and stability of corticosterone in feathers. *J. Exp. Biol.* **212**, 1477–1482. (doi:10.1242/jeb.022152)
65. Kennedy EA, Lattin CR, Romero LM, Dearborn DC. 2013 Feather coloration in museum specimens is related to feather corticosterone. *Behav. Ecol. Sociobiol.* **67**, 341–348. (doi:10.1007/s00265-012-1454-9)
66. Beattie UK, Romero LM. 2023 Long term stability of corticosterone in feathers. *Comp. Biochem. Physiol. A Mol. Integ. Physiol.* **283**, 111472. (doi:10.1016/j.cbpa.2023.111472)
67. Ataallahi M, Nejad JG, Song JI, Kim JS, Park KH. 2020 Effects of feather processing methods on quantity of extracted corticosterone in broiler chickens. *J. Anim. Sci. Technol.* **62**, 884–892. (doi:10.5187/jast.2020.62.6.884)
68. Dillon D, Fernández Ajó A, Hunt KE, Buck CL. 2021 Investigation of keratinase digestion to improve steroid hormone extraction from diverse keratinous tissues. *Gen. Comp. Endocrinol.* **309**, 113795. (doi:10.1016/j.ygcen.2021.113795)
69. R Core Team. 2023 R: a language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing. See <https://www.R-project.org/>.
70. Posit team. 2023 Rstudio: integrated development environment for R. Boston, MA: Posit Software, PBC. See <http://www.posit.co/>
71. Fox J, Weisberg S. 2019 *An R companion to applied regression*, 3rd edn. Thousand Oaks, CA: Sage. See <https://socialsciences.mcmaster.ca/jfox/Books/Companion>.
72. de Mendiburu F, Yaseen M. 2023 *Agricolae*: statistical procedures for agricultural research. See <https://CRAN.R-project.org/package=agricolae>
73. Reynolds M. 2002 *The foraging ecology, habitat use, and population dynamics of the Laysan Teal (Anas Laysanensis)*. Blacksburg, VA: Virginia Polytechnic Institute and State University.
74. Berkowitz P, Storlazzi CD, Courtot KN, Krause CM, Reynolds MH. 2012 Sea-level rise and wave-driven inundation models for Laysan Island. In *Predicting sea-level rise vulnerability of terrestrial habitat and wildlife of the Northwestern Hawaiian Islands* (eds MH Reynolds, P Berkowitz, KN Courtot, CM Krause), pp. 72–126. Reston, VA: U.S. Department of the Interior, U. S. Geological Survey. (doi:10.3133/ofr20121182)
75. Krause CM, Courtot KN, Berkowitz P, Carter J, Reynolds MH. 2012 Climate change vulnerability assessment of the low-lying Northwestern Hawaiian Islands. In *Predicting sea-level rise vulnerability of terrestrial habitat and wildlife of the Northwestern Hawaiian Islands* (eds MH Reynolds, P Berkowitz, KN Courtot, CM Krause), pp. 3–71. Reston, VA: U.S. Department of the Interior, U. S. Geological Survey. (doi:10.3133/ofr20121182)
76. Nishida G, Beardsley J. 2002 A review of the insects and related arthropods of Midway Atoll. *Bishop Museum Occas. Pap.* **68**, 25–69.
77. Brischoux F, Beaugerard E, Mohring B, Parenteau C, Angelier F. 2020 Short-term dehydration influences baseline but not stress-induced corticosterone levels in the house sparrow (*Passer domesticus*). *J. Exp. Biol.* **223**, jeb.216424. (doi:10.1242/jeb.216424)
78. Brusca GA, DeNardo DF, Lourdais O. 2020 Reproductive state and water deprivation increase plasma corticosterone in a capital breeder. *Gen. Comp. Endocrinol.* **288**, 113375. (doi:10.1016/j.ygcen.2019.113375)
79. Kitaysky AS, Wingfield JC, Piatt JF. 1999 Dynamics of food availability, body condition and physiological stress response in breeding Black-Legged Kittiwakes. *Funct. Ecol.* **13**, 577–584. (doi:10.1046/j.1365-2435.1999.00352.x)
80. Kitaysky AS, Piatt JF, Wingfield JC. 2007 Stress hormones link food availability and population processes in seabirds. *Mar. Ecol. Prog. Ser.* **352**, 245–258. (doi:10.3354/meps07074)
81. Barrett RT, Erikstad KE, Sandvik H, Mykssvoll M, Jenni-Eiermann S, Kristensen DL, Moum T, Reiertsen TK, Vikebø F. 2015 The stress hormone corticosterone in a marine top predator reflects short-term changes in food availability. *Ecol. Evol.* **5**, 1306–1317. (doi:10.1002/ece3.1438)
82. Robart AR, Morado MI, Watts HE. 2019 Declining food availability, corticosterone, and migratory response in a nomadic, irruptive migrant. *Horm. Behav.* **110**, 56–67. (doi:10.1016/j.yhbeh.2019.02.007)
83. Wobeser G, Howard J. 1987 Mortality of waterfowl on a hypersaline wetland as a result of salt encrustation. *J. Wildl. Dis.* **23**, 127–134. (doi:10.7589/0090-3558-23.1.127)
84. Jehl JR, Henry AE, Leger JSt. 2012 Waterbird mortality in hypersaline environments: the Wyoming trona ponds. *Hydrobiologia* **697**, 23–29. (doi:10.1007/s10750-012-1167-0)
85. Meteyer CU, Dubielzig RR, Dein FJ, Baeten LA, Moore MK, Jehl JR, Wesenberg K. 1997 Sodium toxicity and pathology associated with exposure of waterfowl to hypersaline playa lakes of southeast New Mexico. *J. VET. Diagn. Invest.* **9**, 269–280. (doi:10.1177/104063879700900308)
86. Dickens MJ, Romero LM. 2013 A consensus endocrine profile for chronically stressed wild animals does not exist. *Gen. Comp. Endocrinol.* **191**, 177–189. (doi:10.1016/j.ygcen.2013.06.014)
87. Cyr NE, Romero LM. 2008 Fecal glucocorticoid metabolites of experimentally stressed captive and free-living starlings: implications for conservation research. *Gen. Comp. Endocrinol.* **158**, 20–28. (doi:10.1016/j.ygcen.2008.05.001)
88. Cyr NE, Romero LM. 2009 Identifying hormonal habituation in field studies of stress. *Gen. Comp. Endocrinol.* **161**, 295–303. (doi:10.1016/j.ygcen.2009.02.001)
89. Dickens MJ, Delehanty DJ, Michael Romero L. 2010 Stress: an inevitable component of animal translocation. *Biol. Conserv.* **143**, 1329–1341. (doi:10.1016/j.biocon.2010.02.032)
90. Jenni L, Keller N, Almasi B, Duplain J, Homberger B, Lanz M, Korner-Nievergelt F, Schaub M, Jenni-Eiermann S. 2015 Transport and release procedures in reintroduction programs: stress and survival in grey partridges. *Anim. Conserv.* **18**, 62–72. (doi:10.1111/acv.12136)
91. Groombridge JJ, Massey JG, Bruch JC, Malcolm TR, Brosius CN, Okada MM, Sparklin B. 2004 Evaluating stress in a Hawaiian honeycreeper, *Paroreomyza montana*, following translocation. *J. Field Ornithol.* **75**, 183–187. (doi:10.1648/0273-8570-75.2.183)
92. Lèche A, Vera Cortez M, Della Costa NS, Navarro JL, Marin RH, Martella MB. 2016 Stress response assessment during translocation of captive-bred Greater Rheas into the wild. *J. Ornithol.* **157**, 599–607. (doi:10.1007/s10336-015-1305-3)
93. Adams NJ, Parker KA, Cockrem JF, Brunton DH, Candy EJ. 2010 Corticosterone responses and post-release survival in translocated North Island Saddlebacks (*Philesturnus rufusater*) in New Zealand. *Emu - Austral. Ornithol.* **110**, 296–301. (doi:10.1071/MU09084)
94. Volfová M, Palme R, Machovcová Z, Voslářová E, Lukešová G, Večerek V. 2022 Translocation stress is reflected in corticosterone metabolites in pheasant (*Phasianus colchicus*) droppings. *Acta Vet. Brno* **91**, 51–58. (doi:10.2754/avb202291010051)
95. Dickens MJ, Delehanty DJ, Romero LM. 2009 Stress and translocation: alterations in the stress physiology of translocated birds. *Proc. R. Soc. B* **276**, 2051–2056. (doi:10.1098/rspb.2008.1778)

96. Teixeira CP, de Azevedo CS, Mendl M, Cipreste CF, Young RJ. 2007 Revisiting translocation and reintroduction programmes: the importance of considering stress. *Anim. Behav.* **73**, 1–13. (doi:10.1016/j.anbehav.2006.06.002)
97. Romero LM. 2004 Physiological stress in ecology: lessons from biomedical research. *Trends Ecol. Evol.* **19**, 249–255. (doi:10.1016/j.tree.2004.03.008)
98. Letty J, Marchandea S, Aubineau J. 2007 Problems encountered by individuals in animal translocations: lessons from field studies. *Ecoscience* **14**, 420–431. (doi:10.2980/1195-6860(2007)14[420:PEBIAJ]2.0.CO;2)
99. Armstrong DP, Seddon PJ. 2008 Directions in reintroduction biology. *Trends Ecol. Evol.* **23**, 20–25. (doi:10.1016/j.tree.2007.10.003)
100. Romero LM. 2002 Seasonal changes in plasma glucocorticoid concentrations in free-living vertebrates. *Gen. Comp. Endocrinol.* **128**, 1–24. (doi:10.1016/s0016-6480(02)00064-3)
101. Bonier F, Moore IT, Martin PR, Robertson RJ. 2009 The relationship between fitness and baseline glucocorticoids in a passerine bird. *Gen. Comp. Endocrinol.* **163**, 208–213. (doi:10.1016/j.ygcen.2008.12.013)
102. Love OP, Madliger CL, Bourgeon S, Semeniuk CAD, Williams TD. 2014 Evidence for baseline glucocorticoids as mediators of reproductive investment in a wild bird. *Gen. Comp. Endocrinol.* **199**, 65–69. (doi:10.1016/j.ygcen.2014.01.001)
103. Crossin GT, Phillips RA, Lattin CR, Romero LM, Williams TD. 2013 Corticosterone mediated costs of reproduction link current to future breeding. *Gen. Comp. Endocrinol.* **193**, 112–120. (doi:10.1016/j.ygcen.2013.07.011)
104. Legagneux P, Harms NJ, Gauthier G, Chastel O, Gilchrist HG, Bortolotti G, Bêty J, Soos C. 2013 Does feather corticosterone reflect individual quality or external stress in arctic-nesting migratory birds? *PLoS One* **8**, e82644. (doi:10.1371/journal.pone.0082644)
105. Fairhurst GD, Champoux L, Hobson KA, Rail JF, Verreault J, Guillemette M, Montevecchi WA, Brousseau P, Soos C. 2017 Feather corticosterone during non-breeding correlates with multiple measures of physiology during subsequent breeding in a migratory seabird. *Compar. Biochem. Physiol. A Molecul. Integrat. Physiol.* **208**, 1–13. (doi:10.1016/j.cbpa.2017.02.024)
106. Escribano-Avila G, Pettorelli N, Virgós E, Lara-Romero C, Lozano J, Barja I, Cuadra FS, Puerta M. 2013 Testing cort-fitness and cort-adaptation hypotheses in a habitat suitability gradient for roe deer. *Acta Oecol.* **53**, 38–48. (doi:10.1016/j.actao.2013.08.003)
107. Kilgour DAV, Romero LM, Reed JM. 2024 Data from: Feather corticosterone is lower in translocated and historical populations of the endangered Laysan duck (*Anas laysanensis*). Dryad (doi:10.5061/dryad.4tmpg4fjg)
108. Kilgour DAV, Romero LM, Reed JM. 2024 Supplementary material from: Feather corticosterone is lower in translocated and historical populations of the endangered Laysan duck (*Anas laysanensis*). Figshare. (doi:10.6084/m9.figshare.c.7204034)